

Linear and Penalized Regressions

MFE5340: AI in FE: Quantitative Investment

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Importance

- A basic, yet very important, tool in quantitative investment is the (multiple) linear regression model. Athough nonlinearity sounds appealing, **linear models remain the most important ones in financial markets**. Recall what Bryan Kelly said regarding nonlinearity?

- Imagine you'are a new quant researcher, and your portfolio manager asks you to analyze which variables can predict the expected returns of different stocks. What do you do? The anwser is cross-sectional regression.

- For example, the well-known BARRA risk factor model estimates its parameters through cross-sectional regression.

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- A linear regression model allows us to examine the relationship between some given variable x_i and y while controlling for other variables.

- In the general case, assuming there are K explanatory variables, the multivariate linear regression model is:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_K x_K + u.$$

- Other names for x and y:

y	x
Explained variable	Explanatory variable
Response variable	Control variable
Predicted variable	Predictor variable
Regressand	Regressor

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Gauss-Markov Assumptions

- **Assumption 1**: The true relationship in the population is linear.
- **Assumption 2**: The sample is randomly drawn from the population.
- **Assumption 3**: There is no perfect multicollinearity among the explanatory variables (including the intercept term, which is a vector of ones).
- **Assumption 4**: The population error term u has a zero conditional mean given all explanatory variables, i.e., $\mathbb{E}[u \mid x_1, x_2, \cdots, x_K] = 0$.
- **Assumption 5**: The error term has constant variance given the explanatory variables (homoskedasticity), i.e., $\text{var}(u \mid x_1, x_2, \cdots, x_K) = \sigma^2$.

Assumptions 1 through 4 ensure that the OLS estimator is **unbiased**. When Assumptions 1 through 5 all hold, the OLS estimator is the **Best Linear Unbiased Estimator (BLUE)** — that is, it has the smallest variance among all linear unbiased estimators.

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- The regression model in matrix form is:

$$\boldsymbol{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{u},$$

where:
  - $\boldsymbol{y}$ be an $N \times 1$ vector of the dependent variable.
  - $\mathbf{X}$ be an $N \times (K+1)$ matrix of the independent variables, where the first column is a vector of ones for the intercept term.
  - $\boldsymbol{\beta}$ be a $(K+1) \times 1$ vector of coefficients.
  - $\boldsymbol{u}$ be an $N \times 1$ vector of the error terms.

- The OLS estimator:

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- Variance-covariance matrix of $\hat{\boldsymbol{\beta}}$:

$$\sigma^2 \left(\mathbf{X}^\top \mathbf{X}\right)^{-1}.$$

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- The mean squared error (MSE) of the OLS estimator is:

$$\begin{aligned}
\text{MSE}(\hat{\boldsymbol{\beta}}) &= \mathbb{E}\left[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})^\top(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})\right] \\
&= \mathbb{E}\left\{\text{tr}\left[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})^\top(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})\right]\right\} \\
&= \mathbb{E}\left\{\text{tr}\left[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})^\top\right]\right\} \\
&= \text{tr}\underbrace{\left\{\mathbb{E}\left[(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta})^\top\right]\right\}}_{\text{cov}(\hat{\boldsymbol{\beta}}),\ \text{assuming } \hat{\boldsymbol{\beta}} \text{ is unbiased}} \\
&= \text{tr}\left[\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right] \\
&= \sigma^2\text{tr}[(\mathbf{X}^\top\mathbf{X})^{-1}].
\end{aligned}$$

- When covariates are highly correlated, $(\mathbf{X}^\top\mathbf{X})^{-1}$ is ill-conditioned and $\text{tr}[(\mathbf{X}^\top\mathbf{X})^{-1}]$ quickly explodes. (And recall our goal is prediction accuracy, rather than parameter adjudication!)

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So, Do We Stick to BLUE?

- $\text{MSE}(\hat{\boldsymbol{\beta}})$ can be decomposed to:
$$\text{MSE}(\hat{\boldsymbol{\beta}}) = \text{Variance} + \text{Bias}^2.$$

- Biased estimators do exist (e.g. shrinkage estimators). The OLS estimator's variance can be too large as OLS coefficients are unregulated. Biased estimators can reduce overall MSE if they produce substantially smaller variance within the set of linear models. (This lecture.)

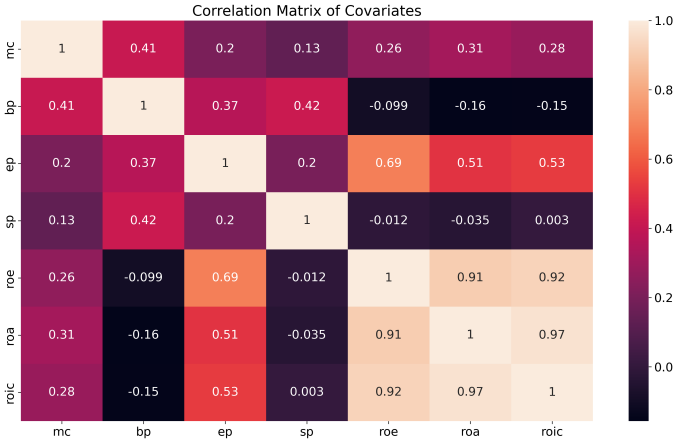
- Likewise, nonlinear estimators do exist (e.g. neural networks). These estimators could be more effective within the set of unbiased models, as they can better capture complex relationships without being constrained by linearity. (Future lectures.)

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Recall the Example

- Independent variables from two categories:
  - **Value**: book-to-market ratio (BP), earnings-to-price ratio (EP), sales-to-price ratio (SP)
  - **Profitability**: return on equity (ROE), return on total asset (ROA), return on invested capital (ROIC)
  - Besides, logarithm of market cap, i.e., ln(market cap).



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	coef	std err	t	P>|t|	[0.025	0.975]
mc	-0.0126	0.007	-1.911	0.057	-0.026	0.000
bp	0.0085	0.008	1.031	0.303	-0.008	0.025
ep	-0.0412	0.010	-4.012	0.000	-0.061	-0.021
sp	0.0045	0.006	0.765	0.445	-0.007	0.016
roe	0.0439	0.019	2.307	0.021	0.007	0.081
roa	0.0435	0.023	1.884	0.060	-0.002	0.089
roic	-0.0423	0.026	-1.654	0.099	-0.093	0.008

Omnibus:	310.897	Durbin-Watson:	1.892
Prob(Omnibus):	0.000	Jarque-Bera (JB):	3665.514
Skew:	2.530	Prob(JB):	0.00
Kurtosis:	15.262	Cond. No.	11.8

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- OLS coefficient (-0.0423) suggests that ROIC is **negatively** correlated to next month returns. This means that, all else being equal, firms with lower profitability measured by ROIC will earn higher expected returns.

- As a matter of fact, the correlation coefficient between ROIC and next month return is 0.117. So, ROIC and return are **positively** correlated.

- Multicollinearity is the key. Since ROIC is highly and positively correlated with ROE and ROA, in OLS, the negative regression coefficient is required to **offset** the positive regression coefficients of ROA and ROE.

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Solution: Penalized Regression

- Penalized regression modifies the OLS objective function by adding a penalty term:

$$\min_{\beta_1,\cdots,\beta_K} \sum_{i=1}^{N}(y_i - \beta_0 - \beta_1 x_{1i} - \cdots - \beta_K x_{Ki})^2 + \lambda \sum_{j=1}^{K} f(\beta_j),$$

  where λ is the regularization parameter. **Note that the intercept β_0 has been left out of the penalty term.**

- OLS does not impose any restriction on the number or size of the regression coefficients, and this can lead to overfitting; Penalized regression can be used to address the problem of overfitting.

- From a Bayesian perspective, this brings *prior knowledge* to the estimation process.

- The penalty term pulls the value of a coefficient closer to zero, reducing its impact on the model. This process is called *shrinkage*. In extreme cases, the penalty term can shrink a coefficient to zero, effectively removing the corresponding covariate from the model (i.e., it achieves *variable selection*).

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- Ridge regression (Hoerl and Kennard 1970) adds a penalty proportional to the sum of the squares of the coefficients (L_2 regularization), i.e., $f(\beta_j) = \beta_j^2$, and the objective function in matrix form becomes (assuming no intercept):

$$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda(\boldsymbol{\beta}^\top \boldsymbol{\beta}).$$

- The estimator:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- The hyperparameter λ constrols for the amount of regularization.

- Including λ makes $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K)$ nonsingular even when $\mathbf{X}^\top \mathbf{X}$ is noninvertible.

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$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- As $\lambda \to 0$, it reduces to OLS estimator; while as $\lambda \to \infty$, $\hat{\boldsymbol{\beta}}^{\text{Ridge}} \to 0$.
- Ridge regression shrinks the coefficients but does not set any of them exactly to zero, retaining all covariates. (**No sparse representation.**)
- Validation is employed to tune λ. The validation set chooses λ that provides the smallest loss function (e.g., MSE).
- The Ridge estimator is **biased**, i.e., $\mathbb{E}[\hat{\boldsymbol{\beta}}^{\text{Ridge}}] \neq \boldsymbol{\beta}$. But does it matter? No, if it produces better performance OOS.

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Interpretation 1: Data Augmentation

- The minimization problem can be formulated as:

$$\sum_{i=1}^{N}(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2 + \sum_{j=1}^{k}(0 - \sqrt{\lambda}\beta_j)^2.$$

- With transformed data:

$$\mathbf{X}_\lambda = \begin{bmatrix} \mathbf{X} \\ \sqrt{\lambda}\mathbf{I}_K \end{bmatrix}, \quad \boldsymbol{y}_\lambda = \begin{bmatrix} \boldsymbol{y} \\ \mathbf{0}_K \end{bmatrix}.$$

- Ridge estimator has the usual OLS estimator's expression:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \left(\mathbf{X}_\lambda^\top \mathbf{X}_\lambda\right)^{-1} \mathbf{X}_\lambda^\top \boldsymbol{y}_\lambda = \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

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Interpretation 2: Bayes Prior

- Suppose the prior on $\boldsymbol{\beta}$ follows multivariate normal:

$$\boldsymbol{\beta} \sim \mathcal{N}\left(\mathbf{0}, \sigma_0^2 \mathbf{I}_K\right).$$

- With normal prior and likelihood, the posterior mean of $\boldsymbol{\beta}$ is:

$$\hat{\boldsymbol{\beta}}^{\text{posterior}} = \left(\mathbf{X}^\top \mathbf{X} + \frac{\sigma^2}{\sigma_0^2} \mathbf{I}_K\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- The hyperparameter λ in Ridge plays the role of σ^2/σ_0^2 in this Bayesian framework, which combines prior views with the likelihood function.

- There is **no bias** from a Bayesian perspective because $\mathbb{E}\left[\hat{\boldsymbol{\beta}}^{\text{Ridge}}\right]$ is the posterior mean of $\boldsymbol{\beta}$.

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Interpretation 3: Eigenvalues and Eigenvectors

- Consider singular value decomposition (SVD):

$$\mathbf{X}_{N\times K} = \mathbf{U}_{N\times K}\mathbf{\Lambda}^{0.5}_{K\times K}\mathbf{V}^{\top}_{K\times K},$$

where:

- $\mathbf{U}_{N\times K} = [\boldsymbol{U}_1, \cdots, \boldsymbol{U}_K]$ satisfies $\mathbf{U}^{\top}\mathbf{U} = \mathbf{I}_K$,
- $\mathbf{V}_{K\times K} = [\boldsymbol{V}_1, \cdots, \boldsymbol{V}_K]$ is orthonormal, and
- $\mathbf{\Lambda}^{0.5} = \begin{bmatrix} \lambda_1^{0.5} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_K^{0.5} \end{bmatrix}$ is an $K \times K$ matrix with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_K$.

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Interpretation 3: Eigenvalues and Eigenvectors

$$\mathbf{X}_{N\times K} = \mathbf{U}_{N\times K}\mathbf{\Lambda}^{0.5}_{K\times K}\mathbf{V}^{\top}_{K\times K}.$$

- $\mathbf{X}^{\top}\mathbf{X} = \mathbf{V}\mathbf{\Lambda}^{0.5}\underbrace{\mathbf{U}^{\top}\mathbf{U}}_{\mathbf{I}}\mathbf{\Lambda}^{0.5}\mathbf{V}^{\top} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\top}$. Therefore, the columns of $\mathbf{V}$ are eigenvectors of $\mathbf{X}^{\top}\mathbf{X}$ and $(\lambda_1, \cdots, \lambda_K)$ are corresponding eigenvalues.
- $\mathbf{X}\mathbf{X}^{\top} = \mathbf{U}\mathbf{\Lambda}^{0.5}\underbrace{\mathbf{V}^{\top}\mathbf{V}}_{\mathbf{I}}\mathbf{\Lambda}^{0.5}\mathbf{U}^{\top} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\top}$. The columns of $\mathbf{U}$ are eigenvectors of $\mathbf{X}\mathbf{X}^{\top}$.
- By retaining only the largest singular values, we can approximate the original matrix with lower rank, achieving **dimension reduction**.

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Interpretation 3: Eigenvalues and Eigenvectors

- Revisit OLS estimator:

$$\begin{aligned}\hat{\boldsymbol{\beta}}^{\text{OLS}} &= \left(\mathbf{X}^\top\mathbf{X}\right)^{-1}\mathbf{X}^\top\boldsymbol{y} \\ &= \left(\mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top\right)^{-1}\mathbf{V}\boldsymbol{\Lambda}^{0.5}\mathbf{U}^\top\boldsymbol{y} \\ &= \mathbf{V}\boldsymbol{\Lambda}^{-1}\mathbf{V}^\top\mathbf{V}\boldsymbol{\Lambda}^{0.5}\mathbf{U}^\top\boldsymbol{y} \\ &= \mathbf{V}\boldsymbol{\Lambda}^{-0.5}\mathbf{U}^\top\boldsymbol{y} \\ &= \mathbf{V}[\text{diag}(\lambda_1^{-0.5},\cdots,\lambda_K^{-0.5})]\mathbf{U}^\top\boldsymbol{y}.\end{aligned}$$

- The fitted values is:

$$\begin{aligned}\hat{\boldsymbol{y}}^{\text{OLS}} &= \mathbf{X}\hat{\boldsymbol{\beta}}^{\text{OLS}} \\ &= \mathbf{U}\boldsymbol{\Lambda}^{0.5}\mathbf{V}^\top\mathbf{V}[\text{diag}(\lambda_1^{-0.5},\cdots,\lambda_K^{-0.5})]\mathbf{U}^\top\boldsymbol{y} \\ &= \mathbf{U}\mathbf{U}^\top\boldsymbol{y} \\ &= \left[\sum_{j=1}^{K}(\boldsymbol{U}_j\boldsymbol{U}_j^\top)\right]\boldsymbol{y}.\end{aligned}$$

- OLS projects $\boldsymbol{y}$ on all the K columns of $\mathbf{U}$.

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Interpretation 3: Eigenvalues and Eigenvectors

- Now, Ridge estimator:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K\right)^{-1} \mathbf{X}^\top \boldsymbol{y}.$$

- Recall $\mathbf{X}^\top \mathbf{X} = \mathbf{V} \boldsymbol{\Lambda} \mathbf{V}^\top$, then $(\mathbf{X}^\top \mathbf{X}) \boldsymbol{V}_j = \lambda_j \boldsymbol{V}_j$.

- Thus,

$$\begin{aligned} \left(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K\right) \boldsymbol{V}_j &= \left(\mathbf{X}^\top \mathbf{X}\right) \boldsymbol{V}_j + \lambda \boldsymbol{V}_j \\ &= \lambda_j \boldsymbol{V}_j + \lambda \boldsymbol{V}_j \\ &= (\lambda_j + \lambda) \boldsymbol{V}_j. \end{aligned}$$

- Therefore, for $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_K)$, columns of $\mathbf{V}$ are still its eigenvectors, while $\lambda_j + \lambda$ is the j-th eigenvalue.

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Interpretation 3: Eigenvalues and Eigenvectors

- By property of linear algebra, the eigen-decomposition of $\left(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_K\right)^{-1}$ is:

$$\left(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_K\right)^{-1}=\mathbf{V}\left[\text{diag}\left(\frac{1}{\lambda_1+\lambda},\cdots,\frac{1}{\lambda_K+\lambda}\right)\right]\mathbf{V}^\top.$$

- Ridge estimator:

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}}=\left(\mathbf{X}^\top\mathbf{X}+\lambda\mathbf{I}_K\right)^{-1}\mathbf{X}^\top\boldsymbol{y}=\mathbf{V}\left[\text{diag}\left(\frac{\lambda_1^{0.5}}{\lambda_1+\lambda},\cdots,\frac{\lambda_K^{0.5}}{\lambda_K+\lambda}\right)\right]\mathbf{U}^\top\boldsymbol{y}.$$

- The fitted values is:

$$\hat{\boldsymbol{y}}^{\text{Ridge}}=\mathbf{X}\hat{\boldsymbol{\beta}}^{\text{Ridge}}=\mathbf{U}\left[\text{diag}\left(\frac{\lambda_1}{\lambda_1+\lambda},\cdots,\frac{\lambda_K}{\lambda_K+\lambda}\right)\right]\mathbf{U}^\top\boldsymbol{y}.$$

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Interpretation 3: Eigenvalues and Eigenvectors

- OLS:

$$\hat{\boldsymbol{y}}^{\text{OLS}} = \mathbf{U}\mathbf{U}^\top \boldsymbol{y}.$$

- Ridge:

$$\hat{\boldsymbol{y}}^{\text{Ridge}} = \mathbf{U}\left[\text{diag}\left(\frac{\lambda_1}{\lambda_1+\lambda}, \cdots, \frac{\lambda_K}{\lambda_K+\lambda}\right)\right]\mathbf{U}^\top \boldsymbol{y}.$$

Ridge regression shrinks the coefficients of all components, but the shrinkage is more pronounced for components with small λ_j.

- Question for you: what is the difference between Ridge and PCA?

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Revisit the Example

- Apply Ridge regression with $\lambda = 20$ (for illustration; no fine-tuning).
- Compared to OLS, Ridge shrinkages all regression coefficients toward zero, e.g.,

Covariate	$\hat{\beta}^{\text{OLS}}$	$\hat{\beta}^{\text{Ridge}}$
ROE	0.0439	0.0234
ROA	0.0435	0.0217
ROIC	−0.0423	−0.0089

- Ridge reduces (in-sample) R^2 compared to OLS (3.0% vs. 6.4%) by shrinking coefficients to improve generalization (i.e., reduce risk of overfitting), often resulting in more reasonable coefficients and better performance on unseen data despite the lower R^2.

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- Lasso (Least Absolute Shrinkage and Selection Operator) regression (Tibshirani 1996) adds a penalty proportional to the sum of the absolute values of the coefficients (L_1 Regularization), i.e., $f(\beta_j) = |\beta_j|$, and the objective function in matrix form becomes (assuming no intercept):

$$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_1.$$

- There is no analytical solution for Lasso estimator.

- Lasso regression can shrink some coefficients to exactly zero, effectively performing variable selection and producing more interpretable models.

- Both Ridge and Lasso have solutions even when $\mathbf{X}^\top \mathbf{X}$ may not be of full rank or ill conditioned.

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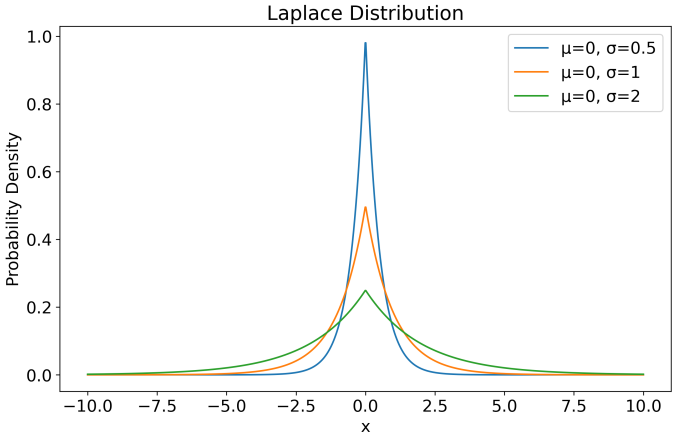
Bayesian Interpretation of Lasso

- Lasso can be interpreted as imposing a **Laplace prior** on the coefficients $\boldsymbol{\beta}$:

$$P(\boldsymbol{\beta}|\lambda) \propto \left(\frac{\lambda}{2\sigma}\right)^K \exp\left(-\frac{\lambda\|\boldsymbol{\beta}\|_1}{\sigma}\right),$$

  where:
  - λ: Controls the dispersion of the prior.
  - σ: Noise standard deviation.

- Laplace distribution has a spike at zero and concentrates probability mass closer to zero.



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- We can view Lasso as a **maximum a posteriori** (MAP) estimator:

$$\hat{\boldsymbol{\beta}}^{\text{Lasso}} = \arg\max_{\boldsymbol{\beta}} p(\boldsymbol{\beta}|\boldsymbol{y}) = \arg\max_{\boldsymbol{\beta}} p(\boldsymbol{y}|\boldsymbol{\beta})p(\boldsymbol{\beta}).$$

- Given the shape of Laplace prior, this leads to sparse solutions, where some coefficients β_j are exactly zero.

- As a comparison, Ridge uses L_2 regularization (Gaussian prior), which does not produce sparsity.

- In contrast, the maximum likelihood estimator would just maximize $p(\boldsymbol{y}|\boldsymbol{\beta})$, without weighting by the prior distribution. In the case of a flat prior, the MAP and maximum likelihood estimator coincide.

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Picking λ

- Same as Ridge, we can still use validation to pick λ.
- However, unlike Ridge, we can also use model selection criteria to pick λ for Lasso because it imposes sparsity.
- Sparsity in Lasso allows explicit calculation of model complexity (number of non-zero coefficients). Thus, BIC or AIC can be used to select λ. For example,

$$\text{BIC} = -2\text{log-likelihood} + K \log(N),$$

where:

— K: Number of non-zero coefficients.
— N: Sample size.

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Why Can't Ridge Use BIC for λ Selection?

- Ridge uses L_2 regularization to shrink coefficients. It does not set any coefficients exactly to zero; all features are retained.
- Without sparsity, the effective number of parameters (K) is unclear. BIC or AIC cannot be directly applied, as K is not well-defined.
- Ridge aims to improve generalization, not feature selection. Validation is more suitable for selecting λ.

Method	**Regularization**	**Sparsity**	**λ Selection**
Lasso	L_1	Yes (Sparse)	BIC, AIC, or Validation
Ridge	L_2	No (Non-Sparse)	Validation

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Revisit the Example (Again)

- Apply Lasso regression with $\lambda = 0.001$ (for illustration; no fine-tuning).
- Compared to OLS, Lasso drops ROIC while shrinking the coefficients for both ROE and ROA:

Covariate	$\hat{\boldsymbol{\beta}}^{\text{OLS}}$	$\hat{\boldsymbol{\beta}}^{\text{Lasso}}$
ROE	0.0439	0.0202
ROA	0.0435	0.0140
ROIC	−0.0423	0

- Similar to Ridge regression, Lasso also reduces the (in-sample) R^2 compared to OLS (2.7% vs. 6.4%).

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- The elastic net is another regularization and variable selection method, where it combines L_1 and L_2 norm penalties:
$$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2.$$
- It still generates sparse representations, making it possible to use model selection criteria for choosing the optimal hyperparameters. (But you can use validation as well!)
- It promotes a grouping effect, where covariates with strong correlations are either included in or excluded from the model collectively.

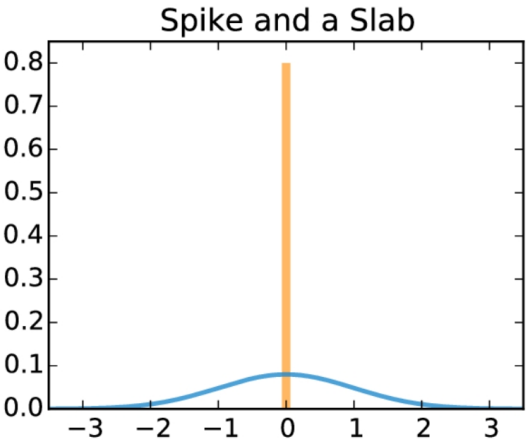
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- In finance, there is large uncertainty about whether the true underlying model is sparse (Lasso) or dense (Ridge).
- The Spike-and-Slab Regression provides a flexible Bayesian framework for **model selection** by encouraging sparsity while allowing non-zero effects.
- We use **binary inclusion indicators** $\boldsymbol{\gamma} = [\gamma_1, \gamma_2, \ldots, \gamma_p]^\top$ where each γ_j:
  - $\gamma_j = 1$: Covariate is included in the model.
  - $\gamma_j = 0$: Covariate is excluded from the model.
- For each γ_j, assume a Bernoulli prior: $\gamma_j \sim \text{Bernoulli}(0.5)$, reflecting an equal probability that each covariate is included or excluded.

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- The **spike** component of the prior concentrates mass at zero for coefficients associated with $\gamma_j = 0$. E.g., Dirac Delta distribution: $\beta_j|\gamma_j = 0 \sim \delta(0)$.
- The **slab** component of the prior distributes mass broadly for coefficients associated with $\gamma_j = 1$, allowing them to take on non-zero values. E.g., a Gaussian distribution can be used: $\beta_j|\gamma_j = 1 \sim \mathcal{N}(0, \sigma_1^2)$, where σ_1^2 is a relatively large variance, enabling the coefficients to vary significantly.
- The posterior distribution of both the inclusion vector $\boldsymbol{\gamma}$ and the coefficients $\boldsymbol{\beta}$ can be estimated by Markov Chain Monte Carlo (MCMC).



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- Model the response variable y_i given covariates $\boldsymbol{x}_i$ and coefficients $\boldsymbol{\beta}$ as:

$$y_i \sim \mathcal{N}(\boldsymbol{x}_i^\top \boldsymbol{\beta}^\star, \sigma_y^2),$$

  where $\boldsymbol{\beta}^\star = \boldsymbol{\gamma} \odot \boldsymbol{\beta}$ (element-wise product), and σ_y^2 is the variance of noise.

- Example: 10 predictors, with the first three covariates related to y.
- True parameter values for relevant covariates: $\boldsymbol{\beta} = [5, -2, 3]^\top$.

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The Spike-and-Slab Regression – Result

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk
gamma[0]	1.000	0.000	1.000	1.000	0.000	0.000	4000.0
gamma[1]	1.000	0.000	1.000	1.000	0.000	0.000	4000.0
gamma[2]	1.000	0.000	1.000	1.000	0.000	0.000	4000.0
gamma[3]	0.018	0.134	0.000	0.000	0.008	0.006	289.0
gamma[4]	0.017	0.129	0.000	0.000	0.003	0.002	1439.0
gamma[5]	0.025	0.155	0.000	0.000	0.011	0.008	202.0
gamma[6]	0.067	0.250	0.000	1.000	0.015	0.011	283.0
gamma[7]	0.012	0.109	0.000	0.000	0.002	0.001	2643.0
gamma[8]	0.010	0.097	0.000	0.000	0.002	0.001	2760.0
gamma[9]	0.012	0.109	0.000	0.000	0.003	0.002	1783.0
beta[0]	4.932	0.112	4.720	5.133	0.002	0.002	2209.0
beta[1]	-2.031	0.094	-2.213	-1.860	0.002	0.001	2070.0
beta[2]	2.940	0.095	2.760	3.115	0.002	0.001	3236.0
beta[3]	-0.067	9.927	-17.657	19.283	0.148	0.201	4509.0
beta[4]	0.356	10.057	-16.858	19.029	0.299	0.256	1233.0
beta[5]	-0.140	9.897	-18.167	19.107	0.199	0.232	2606.0
beta[6]	0.014	9.482	-17.938	17.816	0.135	0.195	4963.0
beta[7]	0.171	9.874	-18.492	18.951	0.156	0.207	3900.0
beta[8]	-0.123	9.851	-19.136	17.437	0.142	0.184	4849.0
beta[9]	-0.057	9.947	-17.707	19.072	0.192	0.204	2663.0

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Ridge, Lasso, and Slack-and-Slab

Method	Regularization	Sparsity	Feature Selection	Prior Distribution
Ridge	L_2	No	No	Gaussian Prior
Lasso	L_1	Yes	Yes	Laplace Prior
Slack and Slab	Mixed Prior	Yes	Yes	Gaussian + Dirac Delta

- **Ridge**: Shrinks coefficients via L_2 regularization but does not produce sparse solutions.
- **Lasso**: Introduces sparsity via L_1 regularization, but treats all coefficients equally (but see next slide!).
- **Slack-and-Slab**: Combines sparsity and non-sparsity, enabling flexible feature selection.

- Lasso forces coefficients to be equally penalized. But we can assign different weights to different coefficients, which leads to the adaptive Lasso:

$$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^{K} w_j |\beta_j|.$$

- Weights w_j can be chosen as $w_j = 1/|\beta_j|^\gamma$, where β_j is the OLS estimate. Why?

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This choice of weights ensures that covariates with larger OLS coefficients receive less penalty, allowing them to retain their influence in the model, while covariates with smaller OLS coefficients are more heavily penalized, encouraging sparsity.

- The adaptive Lasso is a convex optimization problem and thus does not suffer from multiple local minima.

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- In quantitative investment, you can divide covariates (i.e., firm characteristics) into accounting versus market or technical versus fundamental.
- Group Lasso: Divide K covariates into L groups. Model coefficients are $\boldsymbol{\beta} = [\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top, \ldots, \boldsymbol{\beta}_L^\top]^\top$, where $\boldsymbol{\beta}_l$ is the coefficient vector for the l-th group.
- Group Lasso solves the following problem:

$$\mathcal{L}(\boldsymbol{\beta}) = \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right)^\top \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right) + \lambda \left(\sum_{l=1}^{L} \boldsymbol{\beta}_l^\top \boldsymbol{\beta}_l\right)^{1/2}.$$

- Group-Level Selection:
  - If $\|\boldsymbol{\beta}_l\|_2 > 0$, the l-th group is selected, and all variables in the group are retained.
  - If $\|\boldsymbol{\beta}_l\|_2 = 0$, the l-th group is excluded, and all variables in the group are removed.

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The Group Lasso: Within-Group Sparsity

- Group Lasso does not guarantee within-group sparsity.
- If a group is selected, all variables in the group are retained, even if some coefficients are close to zero.
- **Sparse** Group Lasso: Introduce sparsity at both group and variable levels:

$$\mathcal{L}(\boldsymbol{\beta}) = \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right)^\top \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right) + \lambda_1 \left(\sum_{l=1}^{L} \boldsymbol{\beta}_l^\top \boldsymbol{\beta}_l\right)^{1/2} + \lambda_2 \sum_{j=1}^{K} |\beta_j|.$$

- It achieves both group-level and variable-level selections.

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- Bridge regression (Frank and Friedman 1993) generalizes Lasso and Ridge to L_q penalty:

$$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^{K} |\beta_j|^q.$$

  It is immediate clear that $q = 0, 1$, and 2 correspond to OLS, Lasso and Ridge.

- From a machine learning perspective, q is a hyperparameter to be selected, and the solution is sparse for $0 \leq q \leq 1$.

- When the solution is sparse – use model selection criteria to pick hyperparameters (both λ and q). Otherwise, use a validation sample.

Note: This slide incorporates material from Professor Doron Avramov's lecture notes, used with permission.

- Lasso, adaptive Lasso, group Lasso, Ridge, Bridge, and Elastic net are all linear or parametric approaches for shrinkage.

Model	Objective Function
Lasso	$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_1$
Adaptive Lasso	$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^{p} w_j \lvert\beta_j\rvert$
Group Lasso	$\mathcal{L}(\boldsymbol{\beta}) = \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right)^\top \left(\boldsymbol{y} - \sum_{l=1}^{L} \mathbf{X}_l \boldsymbol{\beta}_l\right) + \lambda \left(\sum_{l=1}^{L} \boldsymbol{\beta}_l^\top \boldsymbol{\beta}_l\right)^{1/2}$
Ridge	$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \|\boldsymbol{\beta}\|_2^2$
Bridge	$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \sum_{j=1}^{K} \lvert\beta_j\rvert^q$
Elastic Net	$\mathcal{L}(\boldsymbol{\beta}) = (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda_1 \|\boldsymbol{\beta}\|_1 + \lambda_2 \|\boldsymbol{\beta}\|_2^2$

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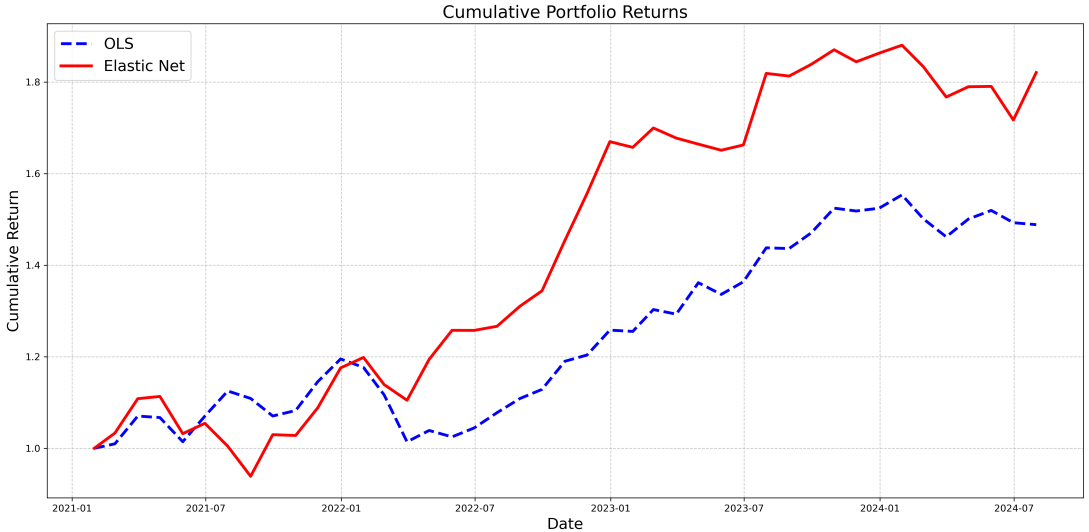


OOS Performance: OLS vs. Elastic Net

- **Covariates**: 54 firm characteristics are used as covariates.
- **Empirical Period**: 2021/1 – 2024/7, monthly frequency.
- **Stock Universe**: Stocks in the CSI 500 Index.
- **Model Training**:
  - Use the past 36 months of data to estimate parameters or train models (OLS and Elastic Net).
  - For Elastic Net, use 5-fold cross-validation to tune hyperparameters.
- **Portfolio Construction**:
  - Predict next month's returns using the trained models.
  - Construct portfolios based on the predicted returns using portfolio sorting.
- **Rolling Window Training**: Repeat the process using a rolling 36-month window to construct out-of-sample portfolios.

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OOS Performance: OLS vs. Elastic Net



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Outliers and Leverage Points

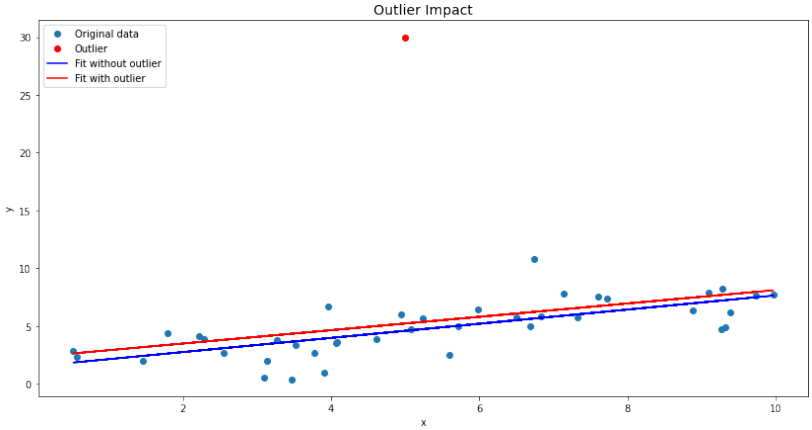
- Outliers are data points that have an unusually large or small response value (y) compared to the other observations. They are located far from the trend of the rest of the data, making them stand out when the data is plotted.
- Outliers can arise from various sources, such as measurement errors, data entry errors, or rare and unique events.
- Leverage points are data points that have an unusual combination of explanatory variable values ($\boldsymbol{x}$). These points are located far from the center of the data distribution in the predictor space, giving them "leverage" over the fitting of the regression model.
- **Not all extreme points necessarily influence the regression model.**

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Example 1: Outlier

5 Robust Regression

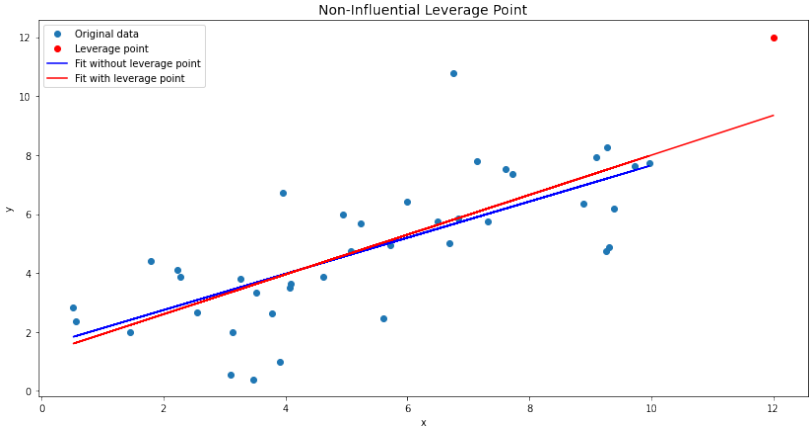
- We generated 30 random data points and added an outlier at $(5, 30)$. The plot shows the best fit line for the original 30 points (in blue) and the line after adding the outlier (in red).
- The outlier shifts the regression line, highlighting its potential influence.



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Example 2: Non-Influential Leverage Point

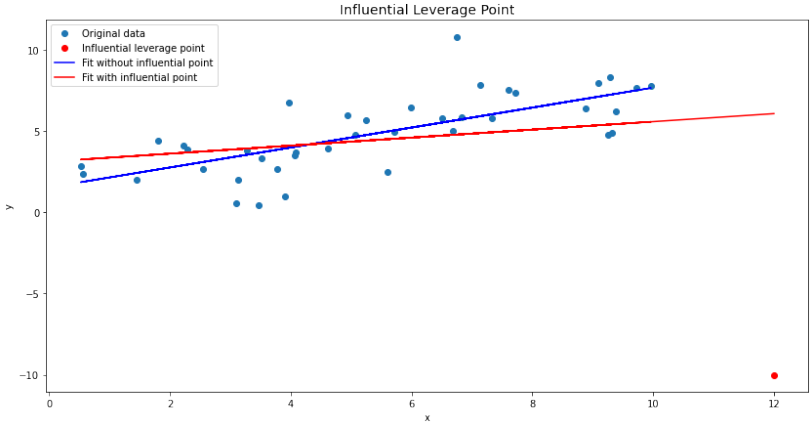
- We added a non-influential leverage point at $(12, 12)$ to the dataset.
- Despite being an extreme value in terms of the x-axis, this point does not significantly change the regression line, as shown by the minimal difference between the blue line and the red line.



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Example 3: Influential Leverage Point

- This example features an influential leverage point at $(12, -10)$.
- This point significantly alters the regression line. Such points can skew the regression results, making it crucial to identify and assess them.



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Cook's Distance

- Cook's Distance (Cook's D) quantifies the influence of individual data points.
- This metric calculates the difference in predicted values *with* and *without* a specific data point in the model.
- For a given observation, its Cook's D is related to the leverage of that observation and its studentized residual.
- A higher Cook's D value indicates a greater influence of the data point on the regression model's estimates.

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- Consider the hat matrix:
$$\mathbf{H} = \mathbf{X}\left(\mathbf{X}^\top \mathbf{X}\right)^{-1}\mathbf{X}^\top.$$
where h_{ii} is the i-th diagonal element of the $\mathbf{H}$.

- The hat value h_{ii} of each observation indicates how far a data point's covariates are from the average covariates in the dataset.

- h_{ii} ranges from 0 to 1. Observations with high leverage are often examined for being potential influential data points.

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- **Studentized** residuals are similar to **standardized** residuals but take into account the variability of the residuals and the leverage of the data points.

- The studentized residual for the i-th observation is:

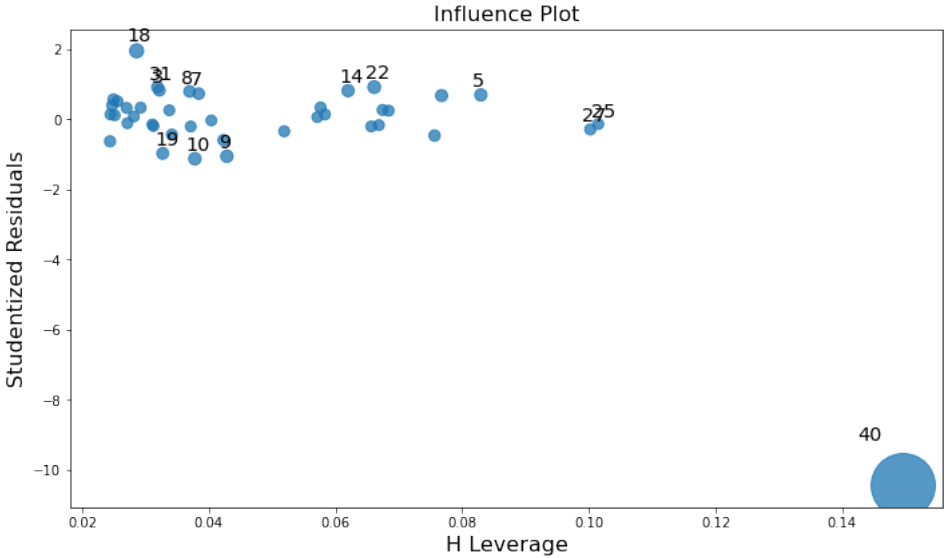
$$\text{sr}_i = \frac{\hat{e}_i}{\text{s.e.}(\hat{e}_i)},$$

  where $\hat{e}_i = y_i - \hat{y}_i$ and $\text{s.e.}(\hat{e}_i)$ is the standard error, i.e., $\text{s.e.}(\hat{e}_i) = \sqrt{\text{MSE} \cdot (1 - h_{ii})}$, with MSE being the mean squared error of the regression model. A studentized residual greater than 2 or less than -2 may suggest an outlier.

- Cook's D:

$$D_i = \frac{1}{K} \times \text{sr}_i^2 \times \frac{h_{ii}}{1 - h_{ii}}.$$

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- Robust regression is a form of regression analysis that can deal with outliers (as well as heteroskedasticity).

- It generalizes the OLS estimator by minimizing the sum of functions of the residuals, rather than the sum of squared residuals:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} \rho(\hat{e}_i),$$

  where $\rho(\hat{e}_i)$ is a function chosen to reduce the influence of outliers.

- First order conditions:

$$\frac{\partial}{\partial \beta_j} \sum_{i=1}^{n} \rho(\hat{e}_i) = \sum_{i=1}^{N} \underbrace{\frac{\partial \rho(\hat{e}_i)}{\partial \hat{e}_i}}_{\psi(\hat{e}_i)} \cdot \underbrace{\frac{\partial \hat{e}_i}{\partial \beta_j}}_{-x_{ij}} = 0.$$

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- Define the weight function $w_i = \frac{\psi(\hat{e}_i)}{\hat{e}_i}$. Then, the first order conditions become $\sum_{i=1}^{N} w_i \hat{e}_i x_{ij} = 0$.
- These conditions appear the same as the first order conditions for Weighted Least Squares (WLS). **Did we just transform robust regression to WLS? NO!**
  - Weights $w_i = \frac{\psi(\hat{e}_i)}{\hat{e}_i}$ depend on residuals $\hat{e}_i$.
  - Residuals $\hat{e}_i$ depend on estimated coefficients, because $\hat{e}_i = y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}_i$.
  - The estimated coefficients $\hat{\boldsymbol{\beta}}$ depend on weights.

It is a circle. We must solve for $\hat{\boldsymbol{\beta}}$ **iteratively**, and this leads to **Iteratively Reweighted Least Squares (IRLS) regression**.

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Iteratively Reweighted Least Squares (IRLS)

1. **Step 1: Initial OLS Estimation**: Start with an initial estimate of $\boldsymbol{\beta}$ using OLS, giving initial residuals $\hat{e}_i$.

2. **Step 2: Compute Weights**: Compute the weights w_i based on the residuals for a chosen ρ function (more on this in the next subsection).

3. **Step 3: Weighted Least Squares**: Perform a WLS regression using the weights w_i to get an updated estimate of $\boldsymbol{\beta}$:
$$\boldsymbol{\beta}^{(\text{new})} = \left(\mathbf{X}^\top \mathbf{W} \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{y}.$$

4. **Step 4: Iterate**: Repeat steps 2 and 3, updating the residuals, weights, and coefficients, until convergence.

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- The ρ function:

$$\rho_{\text{Huber}}(\hat{e}) = \begin{cases} \frac{1}{2}\hat{e}^2 & \text{if } |\hat{e}| \leq k \\ k|\hat{e}| - \frac{1}{2}k^2 & \text{if } |\hat{e}| > k, \end{cases}$$

  where k is a tuning constant.

- **A smaller value of k makes the objective function more resistant to outliers.**

- Typically, k is set to 1.345σ, where σ is the standard deviation of the errors. Use the median absolute deviation (MAD) to estimate σ, i.e., median(|residuals|).

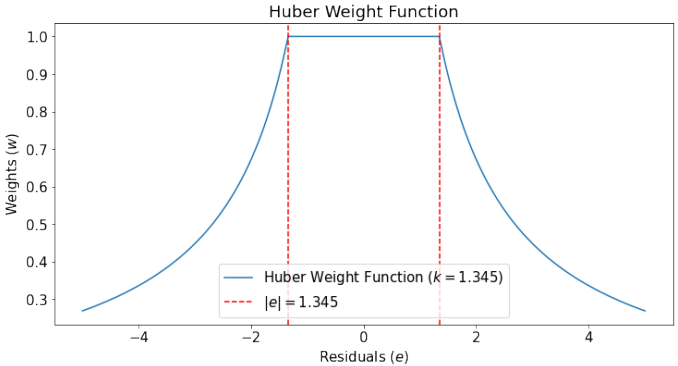
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- The weight function:

$$w_{\text{Huber}}(\hat{e}) = \begin{cases} 1 & \text{if } |\hat{e}| \leq k \\ \dfrac{k}{|\hat{e}|} & \text{if } |\hat{e}| > k. \end{cases}$$

  It assigns a weight of 1 to observations within the threshold k.

- Weights are inversely proportional to the magnitude of the residual for those exceeding the threshold.



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- The ρ function:

$$\rho_{\text{Bisquare}}(\hat{e}) = \begin{cases} \dfrac{k^2}{6}\left[1 - \left(1 - \left(\dfrac{\hat{e}}{k}\right)^2\right)^3\right], & \text{if } |\hat{e}| \leq k, \\ \dfrac{k^2}{6}, & \text{if } |\hat{e}| > k. \end{cases}$$

- k is typically set to 4.685σ; σ again is usually estimated using the MAD to ensure robustness against outliers.

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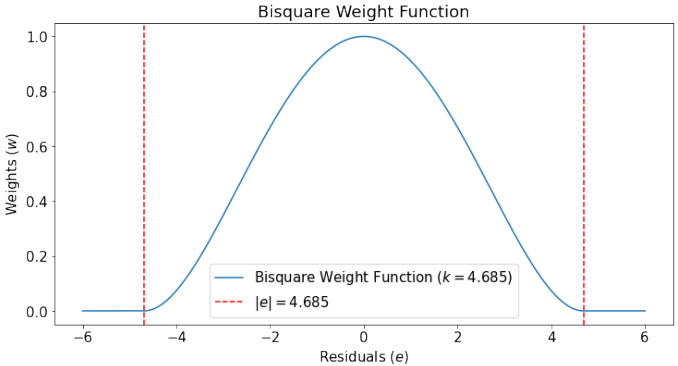
The Bisquare ρ function

- The weight function:

$$w_{\text{Bisquare}}(\hat{e}) = \begin{cases} \left(1 - \left(\frac{\hat{e}}{k}\right)^2\right)^2, & \text{if } |\hat{e}| \leq k, \\ 0, & \text{if } |\hat{e}| > k. \end{cases}$$

  Residuals within the threshold k are assigned weights that decrease smoothly to zero.

- Residuals greater than k are assigned a weight of zero.



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Questions?

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